

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

**Eksamen
VRAESTEL**
Kopiereg voorbehou

2017EBN122E1
Elektrisiteit en Elektronika
November 2017

**Exam
QUESTION PAPER**
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2017EBN122E1
Electricity and Electronics
November 2017



**UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA**

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	93	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 min + 20 min to complete OCR form <i>180 min + 20 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total numer of pages (including this page):</i>		18 + OCR	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word.
All questions must be answered on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form

Interne eksaminator(e): <i>Internal examiner(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye
Interne moderator(e): <i>Internal moderator(s):</i>	J. Joubert

INSTRUCTIONS

Do step 0 before the start of the session.

Do step 1 in the first 180 minutes of the test.

Do step 2 within the following 20 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 180 min):

Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (180 min to 200 min):

Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2017EBN122E1**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Complex numbers:

Rectangular form: place the “*j*” **at the start** of the imaginary part of the complex number.

Polar form: always add the “°” (degree sign) directly after the angle, and make a clear angle sign.

Examples

01	–	1	.	4	+	j	8			μ	A							
02	3	4	0	∠	4	5	°			V								
03	–	j	5	4						k	Ω							
04	1	.	2							r	a	d	÷	s				
05	5	0	∠	–	1	3	0	°		m	A							
06	1	5	5	°														

SECTION / AFDELING 1 [31]

Question 1 to 3 [5]		Vraag 1 to 3 [5]	
01	A smart phone battery is capable of delivering constant current of 2A for 12 hours when fully charged. If the voltage across the terminals of the battery is $10e^{-5t}$ V, where t is in seconds, how much energy is associated with the battery?	01	'n Slimfoonbattery kan 'n konstante stroom van 2A vir 12 ure lewer wanneer dit vol gelaaï is. Indien die spanning oor die battery se terminale $10e^{-5t}$ V, met t wat in sekondes is, hoeveel energie word met die battery geassosieer?
02	It takes 50 minutes to roast a chicken in a 5 kW oven. A restaurant has a budget of R500 per month (30 days) for electricity. How many chickens can they roast before exceeding their budget? (Electricity costs R6/kWh.)	02	Dit vat 50 minute om 'n hoender in 'n 5kW oond te rooster. 'n Restaurant het 'n begroting van R500 per maand (30 dae) vir elektrisiteit. Hoeveel hoenders kan hulle rooster voordat hulle die begroting oorskry? (Elektrisiteit kos R6/kWh.)
03	A voltage source is connected in series with resistors R1, R2, and R3. Calculate the value of R2 if the source supplies 20 W of power to the circuit, R1 and R3 each absorb 7 W of power, and the current through the circuit is 20 mA	03	'n Spanningsbron word in series gekonnekteer met resistors R1, R2 en R3? Bereken die waarde van R2 indien die bron 20 W drywing aan die kringbaan lewer, R1 en R2 elkeen 7 W drywing absorbeer, en die stroom deur die kringbaan 20 mA is.

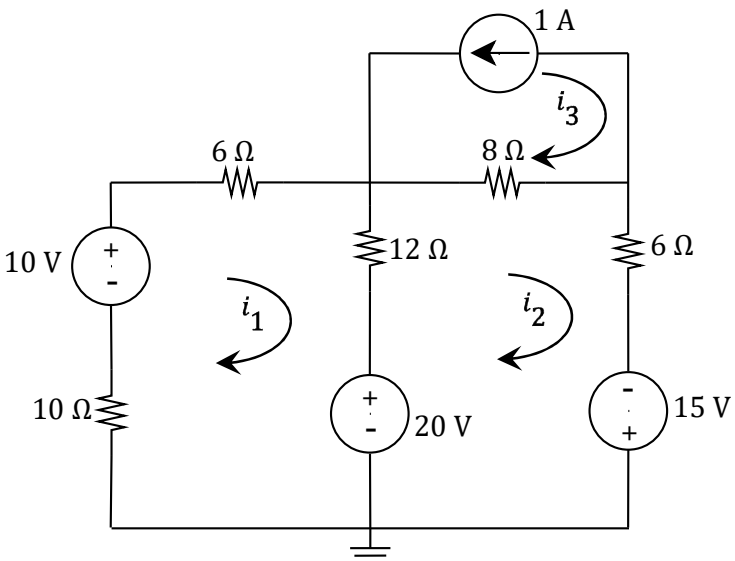
Question 4 & 5 [4]		Vraag 4 & 5 [4]	
Find the power associated with the dependant source and the 6 mA current source. It is given that $V_1 = 5\text{ V}$.		Bereken die drywing geassosieer met die afhanklike bron en die 6 mA stroombron. Dit word gegee dat $V_1 = 5\text{ V}$.	
04	P_d	04	P_d
05	P_{6mA}	05	P_{6mA}

Question 6 & 7 [4]		Vraag 6 & 7 [4]	
06	Find V_I if $V_S = 115\text{ V}$ and $R = 0\ \Omega$.	06	Bepaal V_I indien $V_S = 115\text{ V}$ en $R = 0\ \Omega$.
07	Find i_I if $i_S = 12\text{ mA}$ and $R = \infty\ \Omega$.	07	Bepaal i_I indien $i_S = 12\text{ mA}$ en $R = \infty\ \Omega$.

Question 8 [2]		Vraag 8 [2]	
08	Calculate the equivalent resistance w.r.t. terminals <i>c-d</i>	08	Bereken die ekwivalente weerstand met verwysing tot terminale <i>c-d</i> .

Question 9-12 [4]				Vraag 9-12 [4]			
Use nodal analysis to set up two equations of the following form:				Gebruik node analyse om twee vergelykings van die volgende vorm op te stel:			
$aV_1 + bV_2 = 60$ (KCL equation)				$aV_1 + bV_2 = 60$ (KCL vgl.)			
$cV_1 + dV_2 = 20$ (KVL equation)				$cV_1 + dV_2 = 20$ (KVL vgl.)			
09	a	10	b	09	a	10	b
11	c	12	d	11	c	12	d

Question 13-16 [4]				Vraag 13-16 [4]			
Use mesh analysis to set up two equations of the following form:				Use mesh analysis to set up two equations of the following form:			
$ei_1 - 12i_2 = f$ $-12i_1 + gi_2 = h$				$ei_1 - 12i_2 = f$ $-12i_1 + gi_2 = h$			
13	e	14	f	13	e	14	f
15	g	16	h	15	g	16	h



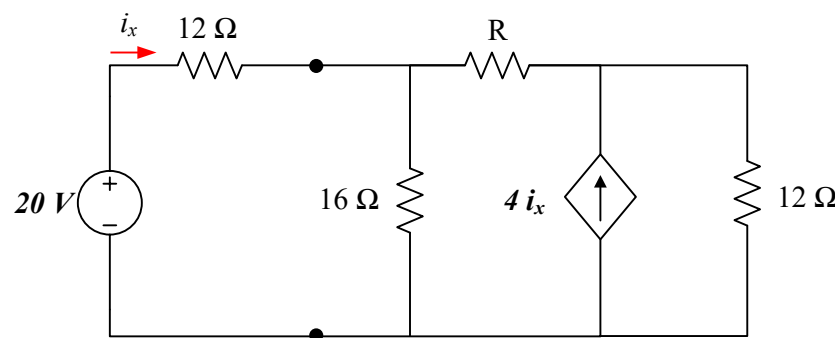
Question 17 [2]		Vraag 17 [2]	
17	Calculate V_0 using either nodal or mesh analysis	17	Bereken V_0 deur gebruik te maak van of node- of lusanalise.

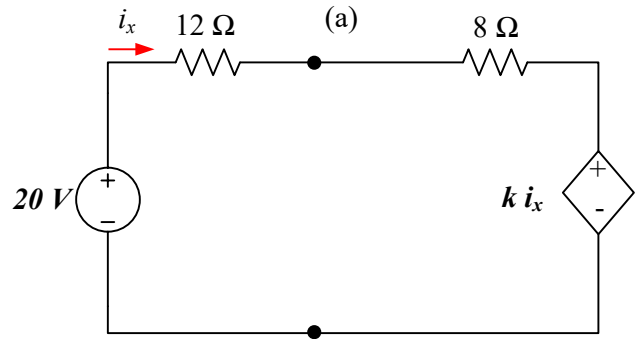
Question 18 [2]		Vraag 18 [2]	
18	Calculate i_0 using either nodal or mesh analysis	18	Bereken i_0 deur gebruik te maak van of node- of lusanalise.

Question 19 & 20 [4]		Vraag 19 & 20 [4]	
Assume that $\beta = 99$ and $V_{BE} = 0.7$ V.		Aanvaar dat $\beta = 99$ en $V_{BE} = 0.7$ V.	
19	Determine V_1 if $R_2 = 3$ k Ω .	19	Bepaal V_1 indien $R_2 = 3$ k Ω .
20	Determine V_2 if $R_2 = 0$ Ω	20	Bepaal V_2 indien $R_2 = 0$ Ω

SECTION / AFDELING 2 [31]

Question 21 & 22 [2]		Vraag 21 & 22 [2]	
Find R and k if Circuit (a) is transformed to Circuit (b).		Bepaal R en k indien Kringbaan (a) na Kringbaan (b) getransformeer word.	
21	R	21	R
22	k	22	k





(a)

(b)

Question 23 & 24 [4]		Vraag 23 & 24 [2]	
1) Find the exclusive contribution of the 3 A current source to i_x .		1) Bepaal die eksklusiewe bydrae van die 3 A stroombron tot die stroom i_x .	
2) Find the exclusive contribution of the 12 V voltage source to i_x .		2) Bepaal die eksklusiewe bydrae van die 12 V spanningsbron tot die stroom i_x .	
23	$i_{x(3A)}$	23	$i_{x(3A)}$
24	$i_{x(12V)}$	24	$i_{x(12V)}$

Question 25 & 26 [4]		Vraag 25 & 26 [4]	
Find the Thevenin equivalent w.r.t. terminals a - b .		Bepaal die Thevenin ekwivalent m.v.t. terminale a - b .	
25	V_{Th}	25	V_{Th}
26	R_{Th}	26	R_{Th}

The circuit diagram shows a 24 V DC voltage source in series with a 3 Ω resistor. This combination is connected to a central node. From this node, a 6 Ω resistor is connected to ground, with the voltage across it labeled v_o . Also from this node, an 8 Ω resistor is connected to terminal a . A dependent current source, labeled $0.5 v_o$ and pointing to the left, is connected between terminal a and terminal b . Terminal b is connected to ground. The terminals a and b are the points of interest for finding the Thevenin equivalent.

Question 27 & 28 [4]		Vraag 27 & 28 [4]	
Find the Norton equivalent w.r.t. terminals a - b .		Bepaal die Norton ekwivalent m.v.t. terminale a - b .	
27	I_N	27	I_N
28	R_N	28	R_N

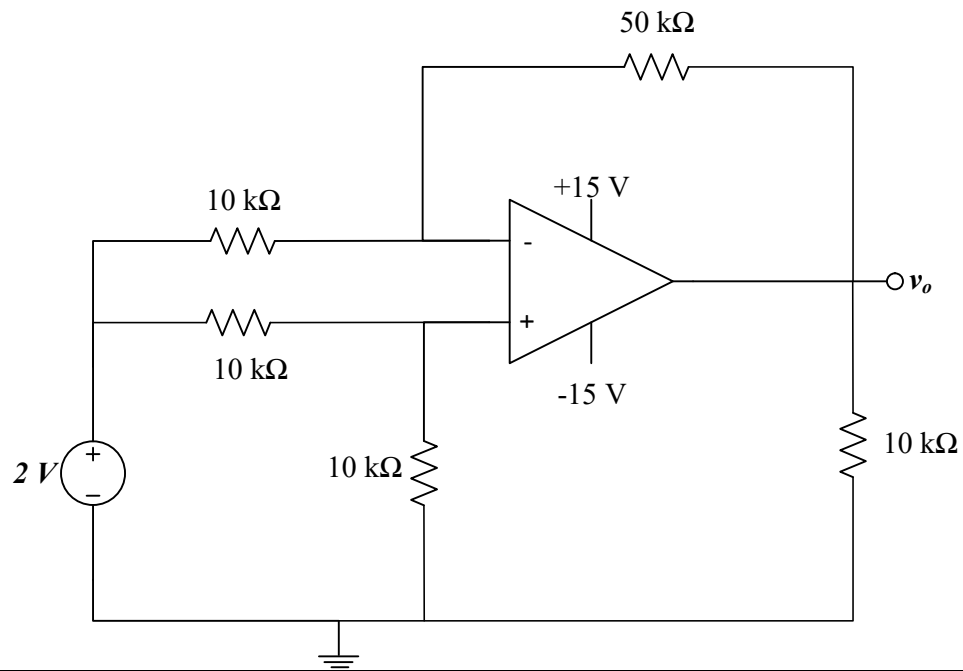
Question 29 & 30 [3]		Vraag 29 & 30 [3]	
1) Calculate the value of the load resistor R_L that can be connected between terminals a - b so that maximum power is transferred to R_L .		1) Bereken die waarde van die lasweerstand R_L wat tussen terminale a - b gekoppel moet word sodat maksimum drywing na R_L oorgedra word.	
2) Determine maximum power P_{max} transferred to R_L .		2) Determine maximum power P_{max} transferred to R_L .	
29	R_L [1]	29	R_L [1]
30	P_{max} [2]	30	P_{max} [2]

Question 31 [2]

Vraag 31 [2]

31 Find v_o .

31 Bepaal v_o .



Question 32 & 33 [4]		Vraag 32 & 33 [4]	
1) Find v_o if $v_1 = 2$ V, $v_2 = 0$ V, and $R_f = 15$ k Ω .		1) Bepaal v_o indien $v_1 = 2$ V, $v_2 = 0$ V, en $R_f = 15$ k Ω .	
2) If $v_2 = 1$ V and $v_1 = 0$ V, then find the maximum value of R_f such that the operational amplifier does not enter a saturation region.		2) Indien $v_2 = 1$ V en $v_1 = 0$ V, bepaal die maksimum waarde van R_f sodat die operasionele versterker nie in 'n versadigingsgebied gaan nie.	
32	v_o	32	v_o
33	R_f	33	R_f

Question 34 & 35 [4]		Vraag 34 & 35 [4]	
1) Find v_o when $v_1 = 0$ V, $v_2 = 2$ V, and $v_3 = 6$ V.		1) Bepaal v_o indien $v_1 = 0$ V, $v_2 = 2$ V, en $v_3 = 6$ V.	
2) Find v_1 when $v_o = -14$ V, $v_2 = 2$ V, and $v_3 = 0$ V.		2) Bepaal v_1 indien $v_o = -14$ V, $v_2 = 2$ V, en $v_3 = 0$ V.	
34	v_o	34	v_o
35	v_1	35	v_1

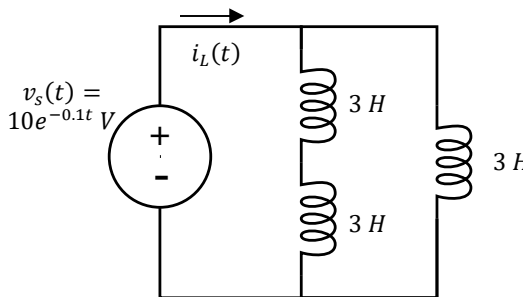
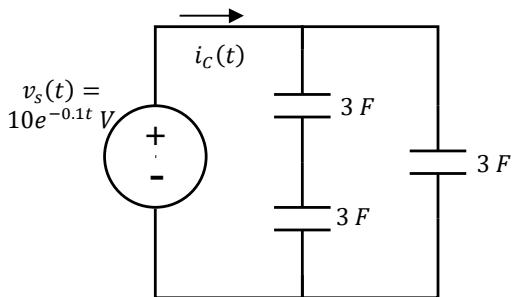
The circuit diagram shows an operational amplifier configured as a summing amplifier. The inverting input (marked with a '-') is connected to three input nodes: v_1 through a $4\text{ k}\Omega$ resistor, v_2 through a $3\text{ k}\Omega$ resistor, and the output v_o through a $12\text{ k}\Omega$ resistor. The non-inverting input (marked with a '+') is connected to v_3 through a $20\text{ k}\Omega$ resistor and to ground through a $40\text{ k}\Omega$ resistor. The op-amp is powered by $+15\text{ V}$ and -15 V supplies. The output v_o is taken from the op-amp's output terminal.

Question 36 & 37 [4]		Vraag 36 & 37 [4]	
1) If $R_f \rightarrow \infty$ and $v_i = 10\text{ mV}$, determine v_o .		1) Indien $R_f \rightarrow \infty$ en $v_i = 10\text{ mV}$, bepaal v_o .	
2) If $R_f = 5\text{ k}\Omega$, determine v_o/v_i .		2) Indien $R_f = 5\text{ k}\Omega$, bepaal v_o/v_i .	
36	v_o	36	v_o
37	v_o/v_i	37	v_o/v_i

The circuit diagram shows two operational amplifiers (op-amps) configured as follows:

- First Op-Amp (Left):** An inverting amplifier. The input signal v_i is applied through a $5\text{ k}\Omega$ resistor to the inverting input (-). The non-inverting input (+) is connected to ground. The feedback resistor is R_f , connected between the output and the inverting input. The output of this stage is labeled v_I . The op-amp is powered by $+15\text{ V}$ and -15 V supplies.
- Second Op-Amp (Right):** A non-inverting amplifier. The input v_I is applied to the non-inverting input (+) through a voltage divider consisting of a $6\text{ k}\Omega$ resistor (connected to v_I) and a $2\text{ k}\Omega$ resistor (connected to ground). The inverting input (-) is connected to ground. A feedback resistor of $5\text{ k}\Omega$ is connected between the output and the inverting input. The output of this stage is labeled v_o . The op-amp is also powered by $+15\text{ V}$ and -15 V supplies.

SECTION / AFDELING 3 [31]

Question 38 & 39 [4]		Vraag 38 & 39 [4]	
At $t = 10$ s, what is the total current through the combination of inductors for Circuit A, and the total current through the combination of capacitors for Circuit B? Assume $v_L(0) = -20$ V, and $v_C(0) = 0$ V.		By $t = 10$ s, wat is die totale stroom deur die kombinasie van induktors vir Kringbaan A, en die totale stroom deur die kombinasie van kapasitors vir Kringbaan B? Aanvaar $v_L(0) = -20$ V, en $v_C(0) = 0$ V.	
38	$i_L(10\text{s})$	38	$i_L(10\text{s})$
39	$i_C(10\text{s})$	39	$i_C(10\text{s})$
 A  B			

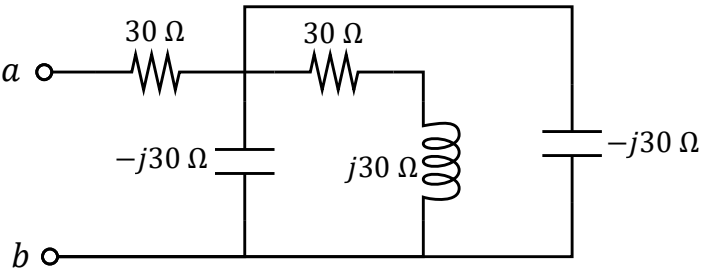
Question 40 & 41 [4]		Vraag 40 & 41 [4]	
Assume steady-state conditions (Direct Current) for the circuit below. What is the energy stored by the capacitor w_C and inductor w_L ?		Aanvaar gestadigde toestand (Gelykstroom) vir onderstaande kringbaan. Wat is die energie gestoor deur die kapasitor w_C en induktor w_L ?	
40	w_C	40	w_C
41	w_L	41	w_L

Questions 42 & 43 [4]		Vraag 42 & 43 [4]	
42	The following sinusoid is given: $v_1 = 8 \cos(3t - 45^\circ) \text{ V}$ It is known that: - v_1 leads v_2 by 160°. - v_1 lags v_3 by 150°. What is the phase angle θ between v_2 and v_3? (Give the angle in the range of -180° to 180°.)	42	Die volgende sinusied word gegee: $v_1 = 8 \cos(3t - 45^\circ) \text{ V}$ Dit is bekend dat: - v_1 loop voor v_2 met 160°. - v_1 loop agter v_3 met 150°. Wat is die fasehoek θ tussen v_2 en v_3? (Gee die hoek in die bereik van -180° tot 180°.)
43	The following sinusoids are given: $i_1 = -1.766 \sin(\omega t - 30^\circ) \text{ A}$ $i_2 = -2 \cos(\omega t - 45.4^\circ) \text{ A}$ Determine the phasor value of $\bar{I}_T = \bar{I}_1 + \bar{I}_2$. Give the answer in polar form.	43	Die volgende sinusiede word gegee: $i_1 = -1.766 \sin(\omega t - 30^\circ) \text{ A}$ $i_2 = -2 \cos(\omega t - 45.4^\circ) \text{ A}$ Bepaal die fasor waarde van $\bar{I}_T = \bar{I}_1 + \bar{I}_2$. Gee die antwoord in polêre vorm.

Question 44 [2]		Vraag 44 [2]	
The following four elements are all connected in parallel: a $10 \cos(2000t + 60^\circ) \text{ A}$ current source, a 4Ω resistor, a $125 \mu\text{F}$ capacitor and a 2 mH inductor. What is the voltage across the resistor? (Assume the positive form of Ohm's law. Give the answer in polar form .)		Die volgende vier elemente word in parallel gekoppel: 'n $10 \cos(2000t + 60^\circ) \text{ A}$ stroombron, 'n 4Ω resistor, 'n $125 \mu\text{F}$ kapasitor en 'n 2 mH induktor. Wat is die spanning oor die resistor? (Aanvaar die positiewe vorm van Ohm se Wet. Skryf die antwoord in polêre vorm .)	
44	$\bar{V}_R$	44	$\bar{V}_R$

Question 45 & 46 [4]		Vraag 45 & 46 [4]	
Determine $\bar{I}_T$ and $\bar{V}_{ab}$.		Bepaal $\bar{I}_T$ en $\bar{V}_{ab}$.	
45	$\bar{I}_T$ (rectangular form.)	45	$\bar{I}_T$ (rectangular form.)
46	$\bar{V}_{ab}$ (rectangular form.)	46	$\bar{V}_{ab}$ (rectangular form.)

The circuit diagram shows a 500 V AC voltage source on the left. A branch in parallel with the source contains a $20\ \Omega$ resistor and a $-j15\ \Omega$ capacitor. The current through this branch is labeled $\bar{I}_T$. To the right of this branch is another parallel branch containing a $j3\ \Omega$ inductor and a $4\ \Omega$ resistor. The voltage across this branch is labeled $\bar{V}_{ab}$, with terminal 'a' at the top and terminal 'b' at the bottom. The bottom wire is grounded.

Question 47 [2]		Vraag 47 [2]	
47	Calculate Z_{ab} . (Give the answer in rectangular form .)	47	Bereken Z_{ab} . (Skryf die antwoord in reghoekige vorm .)
 The circuit diagram shows terminals a and b. A $30\ \Omega$ resistor is connected to terminal a. After this resistor, the circuit splits into two parallel branches. The first branch contains a $-j30\ \Omega$ capacitor. The second branch contains a $30\ \Omega$ resistor in series with a $j30\ \Omega$ inductor. These two branches recombine, and the circuit then splits again into two parallel branches. The first branch contains a $-j30\ \Omega$ capacitor. The second branch is a direct wire connecting the top and bottom lines. Finally, the circuit connects to terminal b.			

Question 48 [2]		Vraag 48 [2]	
48	Determine $\bar{V}_0$ as a result of the $2\angle 90^\circ V$ voltage source alone. Use source transformation (Write the answer in rectangular form .)	48	Bepaal $\bar{V}_0$ slegs as gevolg van die $2\angle 90^\circ V$ spanningsbron. Gebruik brontransformasie. (Skryf die antwoord in reghoekige vorm .)

Question 49 to 52 [4]				Vraag 49 to 52 [4]			
Use nodal analysis to set up two equations of the following form $a\bar{V}_1 + b\bar{V}_2 = 2$ $c\bar{V}_1 + d\bar{V}_2 = 8 + j6$ Give the coefficients in rectangular form .				Gebruik node analyse om twee vergelykings van die volgende vorm op te stel: $a\bar{V}_1 + b\bar{V}_2 = 2$ $c\bar{V}_1 + d\bar{V}_2 = 8 + j6$ Gee die koëffisiënte in reghoekige vorm .			
49	a	50	b	49	a	50	b
51	c	52	d	51	c	52	d

Question 53 to 56 [4]				Vraag 53 to 56 [4]			
Use nodal analysis to set up two equations of the following form $e\bar{I}_1 + f\bar{I}_2 = 5\angle -120^\circ$ $g\bar{I}_1 + h\bar{I}_2 = 2$ Give the coefficients in rectangular form .				Gebruik node analyse om twee vergelykings van die volgende vorm op te stel: $e\bar{I}_1 + f\bar{I}_2 = 5\angle -120^\circ$ $g\bar{I}_1 + h\bar{I}_2 = 2$ Gee die koëffisiënte in reghoekige vorm .			
53	e	54	f	53	e	54	f
55	g	56	h	55	g	56	h

Question 56 & 57 [4]		Vraag 56 & 57 [4]	
Determine the Thevenin equivalent w.r.t. terminals a - b .		Bepaal die Thevenin ekwivalent m.v.t. terminale a - b .	
56	$\bar{V}_{TH}$ (rectangular form.)	56	$\bar{V}_{TH}$ (rectangular form.)
57	Z_{TH} (rectangular form.)	57	Z_{TH} (rectangular form.)

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PRACTICE PAGE / OEFENBLAD 1

Assessment ID	2	0	1	7	E	B	N	1	2	2	E	1		
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Make use of **5 significant digits** for all answers involving **irrational numbers**.
*Maak gebruik van **5 beduidende syfers** vir alle antwoorde bestaande uit **irrasionele getalle**.*

Section 1 / Afdeling 1 [31]		
01	$E =$	
02	# of chickens =	No unit required
03	$R_2 =$	
04	$P_d =$	
05	$P_{6mA} =$	
06	$V_I =$	
07	$i_I =$	
08	$R_{cd} =$	
09	$a =$	No Unit Required
10	$b =$	
11	$c =$	
12	$d =$	
13	$e =$	
14	$f =$	
15	$g =$	
16	$h =$	
17	$V_0 =$	
18	$i_0 =$	
19	$V_I =$	
20	$V_2 =$	

PRACTICE PAGE / OEFENBLAD 2

Assessment ID	2	0	1	7	E	B	N	1	2	2	E	1		
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Make use of 5 significant digits for all answers involving irrational numbers.
Maak gebruik van 5 beduidende syfers vir alle antwoorde bestaande uit irrasionele getalle.

Section 2 / Afdeling 2 [31]		
21	$R =$	
22	$k =$	<i>Unit not required</i>
23	$i_{x(3A)} =$	
24	$i_{x(80V)} =$	
25	$V_{Th} =$	
26	$R_{Th} =$	
27	$I_N =$	
28	$R_N =$	
29	$R_L =$	
30	$P_{max} =$	
31	$v_o =$	
32	$v_o =$	
33	$R_f =$	
34	$v_1 =$	
35	$v_o =$	
36	$v_o =$	
37	$v_o/v_i =$	<i>Unit not required</i>

PRACTICE PAGE / OEFENBLAD 3

Assessment ID

2	0	1	7	E	B	N	1	2	2	E	1		
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Remember:

Write a complex number in:

- rectangular form as $a \pm j b$ where j is at the **start** of the imaginary number.
- polar form as $a \angle \theta^\circ$ where the angle sign $\angle$ is clear, and you add the $^\circ$ symbol directly after the angle).

(See examples on page 2.)

Section 3 / Afdeling 3 [31]		
38	$i_L (10s) =$	
39	$i_C (10s) =$	
40	$w_C =$	
41	$w_L =$	
42	$\theta =$	
43	$\bar{I}_T =$ (polar)	
44	$\bar{V}_R =$ (polar)	
45	$\bar{I}_T =$ (rectangular)	
46	$\bar{V}_{ab} =$ (rectangular)	
47	$Z_{ab} =$ (rectangular)	
48	$\bar{V}_0 =$ (rectangular)	
49	$a =$ (rectangular)	No unit required
50	$b =$ (rectangular)	
51	$c =$ (rectangular)	
52	$d =$ (rectangular)	
53	$e =$ (rectangular)	
54	$f =$ (rectangular)	
55	$g =$ (rectangular)	
56	$h =$ (rectangular)	
57	$\bar{V}_{TH} =$ (rectangular)	
58	$Z_{TH} =$ (rectangular)	